

Experiment-5: Subword Level Tokenization, POS Tagging and Word2Vec

5.1. Implement subword-level tokenization using a BPE-based tokenizer. Given input data:

https://drive.google.com/file/d/1eHA0Q9ju-08n_hByCK03GqWAX9HQvAKd/view?usp=sharing

1. Tokenize the sentence into subword units.
2. Display the generated tokens and their corresponding token IDs.

(Implement two separate codes: a. Using pretrained model and b. Without using pretrained model)

5.2. Implement subword-level tokenization using a SentencePiece tokenizer. Given same input data of 5.1

https://drive.google.com/file/d/1eHA0Q9ju-08n_hByCK03GqWAX9HQvAKd/view?usp=sharing

1. Tokenize the sentence into subword units.
2. Display the generated tokens and their corresponding token IDs.

(Implement two separate codes: a. Using pretrained model and b. Without using pretrained model)

5.3. Implement Part-of-Speech (POS) tagging on a given text and display the unique token, POS tag, and description of each POS tag.

"The young student is reading an interesting book in the library."

(Implement two separate codes: a. Using Spacy and b. NLTK)

5.4. Implement Part-of-Speech (POS) tagging on a given text and display the unique token, POS tag, description of each token and frequency using Spacy.

Input data: Same input data of 5.1

https://drive.google.com/file/d/1eHA0Q9ju-08n_hByCK03GqWAX9HQvAKd/view?usp=sharing

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5.5 Write a Python program to train a Word2Vec CBOW model using the given text corpus. Display the vector representation of a selected word and find the three most similar words.

Input corpus

Create a file named `corpus.txt`:

Natural language processing is interesting
Natural language processing is useful
I love machine learning
Machine learning is useful
Deep learning is a part of machine learning
Artificial intelligence is changing the world
Natural language processing uses machine learning
Deep learning and artificial intelligence are related
I study natural language processing
Machine learning helps artificial intelligence

5.6 Write a Python program to train a Skip-Gram Word2Vec model using the same corpus. Retrieve the vector for "language" and display the five most similar words.

Input corpus Use `corpus.txt` from Question 5.5.

5.7 Train a Word2Vec model using the provided corpus and calculate the cosine similarity between selected word pairs.

Input corpus: Use `corpus.txt` from Question 5.5.

Calculate similarity for:

1. machine and learning
2. natural and language
3. deep and learning